MOON MAPPING

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INTRODUCTION

- Chandrayaan 2 had two camera modules, OHRC and TMC.
- OHRC had spatial resolution of 30 cm, whereas the TMC module had spatial resolution of 5 m.
- But OHRC could only click images in some particular light conditions.
- TMC has captured images of more than 50 percent of the lunar surface.
- OHRC and TMC have some overlapping regions.
- The aim of the project is to create an AI/ML model which will upscale the images captured by TMC to 30 cm spatial resolution.
- The further aim is to create a lunar map by stitching the images to create a high resolution lunar atlas.

TASKS

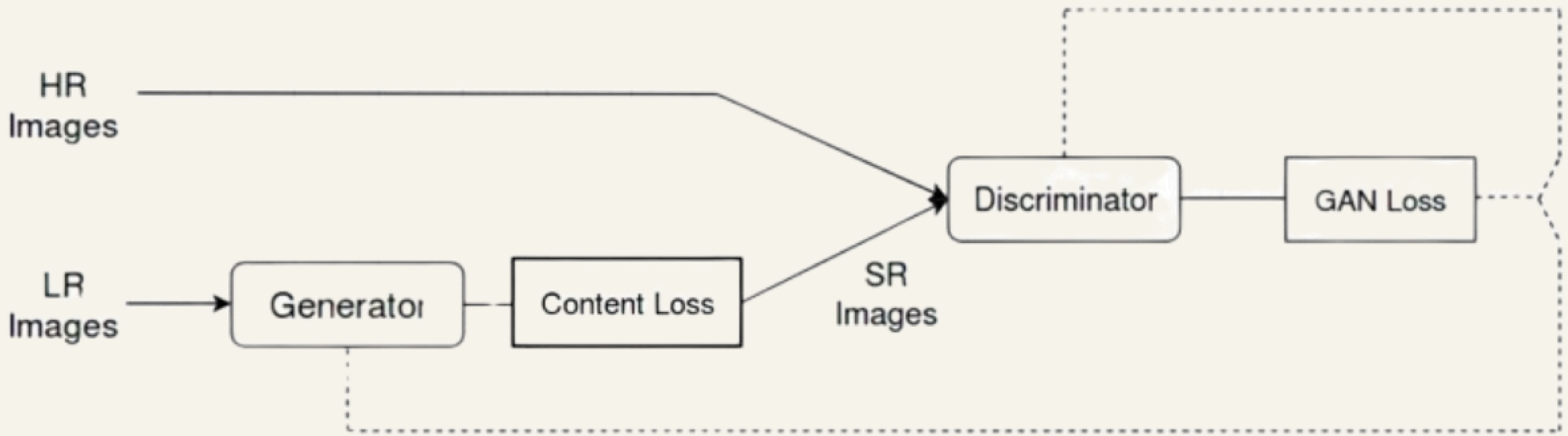
- Collect all the image data of TMC and OHRC
- Preprocess it and link the images capturing same area of moon together.
- Apply Deep Learning models to upscale images from TMC.
- Stich those upscaled images together to get a high res lunar map of moon.

INTRO TO DL MODELS

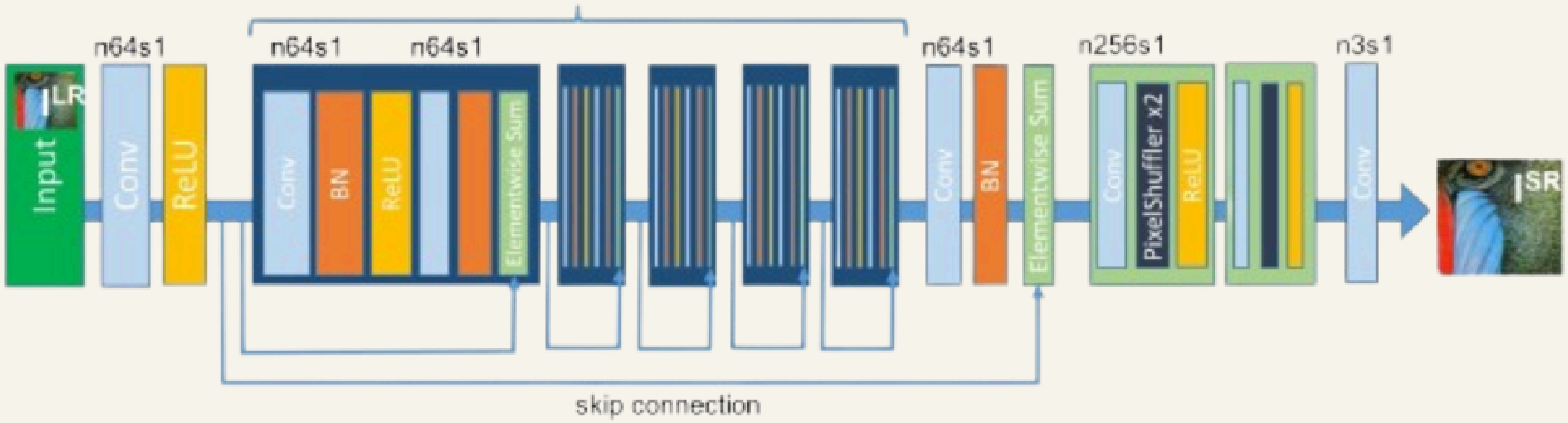
Two Deep Learning models were used to accomplish the upscaling task:

- **SuperRes Generative Adversarial Network (SRGAN)**
- **Hierarchical Vision Transformer using Shifted Windows (SWIN Transformer)**

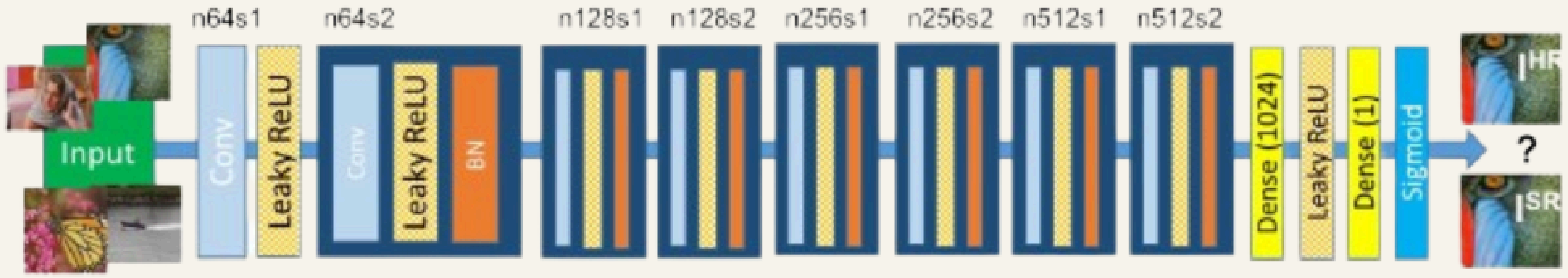
SRGAN


HR
Images
LR
Images
Generator
Content Loss
SR
Images
Discriminator
GAN Loss

DETAILED ARCH



Discriminator Network



DEPLOYEMENT OF GAN